Hasti KabirSamadi

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Amirkabir University

Profile

I am a Computer Engineering senior at Amirkabir University of Technology (Tehran Polytechnic), and biology has steadily become the lens through which I see every machine learning problem I touch. What pulls me in is the idea that the same transformer architectures powering language models might already *know* something deep about RNA structure and function, and that we can surface that knowledge through careful probing without any fine-tuning at all. That is exactly what I am working on right now under Prof. Fatemeh Zare-Mirakabad: developing methods that extract structural and functional predictions directly from the attention matrices and embedding representations of RNA language models. Before this, I built scArches-Plus, a transfer-learning framework for single-cell atlas mapping that cut batch effects by 22% and training time by 40% through architectural redesign. I work daily in Python with PyTorch, Scanpy, and AnnData, and I am actively building my expertise in RNA-seq pipelines and statistical genomics. I am looking for a research internship in computational biology or bioinformatics where I can contribute to real ongoing work, keep learning fast, and grow as a scientist.

Education

B.Sc. in Computer Engineering 2022 – present
Amirkabir University of Technology (Tehran Polytechnic) | Iran
GPA (last three semesters): 18.5 / 20

Farzanegan 2 High School 2017 – 2022
National Organization for Development of Exceptional Talents of Iran

Honors & Awards

Ranked 71st in the nationwide Iranian Universities Entrance Examination among more than 145,000 participants 2022

Research Experience

Research Assistant, Amirkabir University of Technology 2026 – present
Supervisor: Prof. Fatemeh Zare-Mirakabad (Dept. of Mathematics & Computer Science)

- Evaluating RNA language models (RiNALMo, RNA-FM, and related architectures) across structural, functional, and evolutionary tasks in a zero-shot setting.
- Investigating structural and functional signals encoded in transformer attention matrices and contextual embedding representations of RNA sequences.
- Developing a novel probing method that extracts task-relevant predictions solely from attention matrices and embedding matrices — without fine-tuning or downstream task-specific heads.

Research Intern, Iran Telecommunication Research Center (ITRC) Summer 2025

- Investigated multilingual transformer models for Persian sentiment classification.
- Fine-tuned XLM-RoBERTa on Persian Twitter datasets and evaluated performance.
- Conducted error analysis, model comparison, and documented results in a reproducible pipeline.

Academic Projects

scArches-Plus: Transfer-Learning Framework with Correlation Penalty for Single-Cell Atlas Mapping 2025
Bioinformatics course project, Amirkabir University of Technology

- Re-implemented scArches with a Correlation Penalty regulariser → clustering purity +8%.

- Added ExtEncoder layer & redesigned Decoder → batch-effect correction -22% (Silhouette score +0.11).
- Enabled transfer-learning fine-tuning without raw reference data → training time -40%.

Research Interests

- **Bioinformatics & Computational Biology:**
  Single-cell and bulk RNA-seq analysis, gene regulatory network inference, multi-omics data integration, and ML-driven approaches to genomics and healthcare data.
- **Deep Learning for Biological Sequences:**
  Applying transformer and foundation model architectures to DNA, RNA, and protein sequence analysis; probing attention and embedding representations; transfer learning for biological atlas mapping.
- **Natural Language Processing:**
  Fine-tuning and evaluating transformer-based models for sequence classification, covering the full pipeline from dataset preparation and tokenization to supervised fine-tuning and evaluation.

Related Courses and Workshops

Advanced Bioinformatics — in progress
Sharif University of Technology (online)

RNA-seq Data Analysis Workshop — 2026
Genome Guide (online)

Fundamentals of Bioinformatics — 2024
Amirkabir University of Technology

Introduction to NLP: Basic Concepts, Preprocessing and Representations — 2023
Amirkabir University of Technology

Skills & Languages

- **Programming:** Python, C, C++, Java.
- **ML & Data:** PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib.
- **Bioinformatics:** Scanpy, AnnData, scArches; scRNA-seq data processing; RNA-seq analysis pipelines.
- **Tools & Platforms:** Git & GitHub, Docker, Linux/Bash, LaTeX.
- **Languages:** Persian (Native), English (Advanced).

Teaching Experience

Head Teaching Assistant, Microprocessors & Assembly Language (Undergraduate) — Fall 2025

- Amirkabir University of Technology
- Lead and coordinated the TA team: assigned tasks, distributed responsibilities, ensured consistent grading standards.
- Planned assessments (assignments, quizzes, projects) and oversaw QA of questions, model answers, and rubrics.
- Served as primary intermediary between students and the instructor.

Teaching Assistant — Amirkabir University of Technology

- Computer Networks (Spring 2026) · Software Test (Spring 2026) · Software Engineering 2 (Fall 2025)
- Microprocessors & Assembly Language (Spring 2024) · Advanced Programming (Fall 2024, Fall 2025)
- Differential Equations (Spring 2023)